

URBAN ECONOMICS • CASE STUDY

Lubumbashi

*How mining rents shape urban form
in a resource-dependent city*

Abdul Sande • with Matthew & Oge

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The Lubumbashi pattern, in one slide

~75%

of Lubumbashi's workforce is employed in non-tradable sectors — the signature of a 'Consumption City.'

3-5 m

median building height in the densest neighborhoods — a horizontal, low-verticality urbanism.

~74%

of global cobalt supply originates in the DRC; Lubumbashi is its administrative & logistics core.

99%

of fenceline fishers & farmers in nearby communities report sharp livelihood collapse from mining pollution.

Bottom line: A century of mining specialization locked Lubumbashi into a low-density, segmented urban form — with predictable economic, spatial, and human costs.

Lubumbashi — the heart of the DRC Copperbelt



Sub-Saharan Africa: Lubumbashi sits in the south-eastern DRC, 30 km from the Zambian border.

Population	~2.6 M (2nd-largest city in DRC)
Density	~3,000 people / km ²
Founded	1910 as Élisabethville (Belgian colonial)
Role	Capital of Haut-Katanga; copper-cobalt admin hub
Global stake	DRC supplies ~74% of cobalt, >11% of copper
Mines nearby	Tenke Fungurume (~190 km), Kolwezi (~240 km)

The research question



How do mining rents shape urban form in resource-dependent cities like Lubumbashi?

THEORY

Consumption-city, urban-inequality & resource-urbanism literature

EVIDENCE

Employment data, satellite imagery, building heights, wage surveys

OUTCOMES

Urban form, inequality, environmental & human-development indicators

Resource booms create 'Consumption Cities'

Mechanism

In resource-dependent economies, two forces re-allocate labor away from manufacturing and agriculture toward non-tradable services:

1

Income effect

Resource rents raise wages → urban migration from rural areas.

2

Productivity in tradables

Capital-intensive extraction needs few workers → released labor shifts to non-tradable services.

Result

A 'Consumption City'

Cities urbanize without industrializing: they produce mostly non-tradables and use commodity rents to import the tradables they consume.

Predicted form:

- Horizontal sprawl
- Informal settlements
- Weak manufacturing base

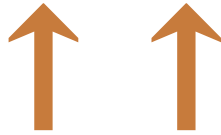
Source: Jedwab, Gollin & Vollrath (2013), *Urbanization with and without Structural Transformation*.

Mining lifts averages, but widens local gaps



Higher consumption

Producing districts see higher household consumption and lower poverty rates than peers.



Wider inequality

Within those same districts, inequality rises sharply — gains accrue to a more-educated, often migrant, workforce.



No spillovers

Benefits attenuate with distance from mines; neighboring districts see little to no gain.

Why governance matters: commodity booms can raise local means while deepening spatial inequality — unless rents are channeled into city-wide public goods (infrastructure, services, skills).

Sources: Loayza & Rigolini (2016); Rubbers (2020).

Form follows resources — and form decides outcomes

Cross-country evidence shows resource endowments shape urban morphology, and morphology — density, land-use mix, transport supply — is what actually drives resource efficiency and equity outcomes.

Land-scarce coastal hubs

Hong Kong · Singapore

- Compact, high-rise core
- Transit-oriented; metro + rail-property integration
- Mixed land use, walkable
- Low per-capita transport energy

Resource-abundant cities

Kuwait · Abu Dhabi

- Low-density villas, sprawling
- Highway-led, car-centric
- Spatially segregated by citizenship/income
- High cooling + transport energy

Bottom line: Resource endowments set the means of urbanization; morphology is the proximate driver of long-run outcomes.

Sources: Rode et al. (2017), LSE Cities · Resource Urbanisms (Bertaud and others, 2018).

What we should see if the theory holds in Lubumbashi

H1

A workforce concentrated in non-tradables, with imports dominating tradable consumption — the 'Consumption-City' signature.

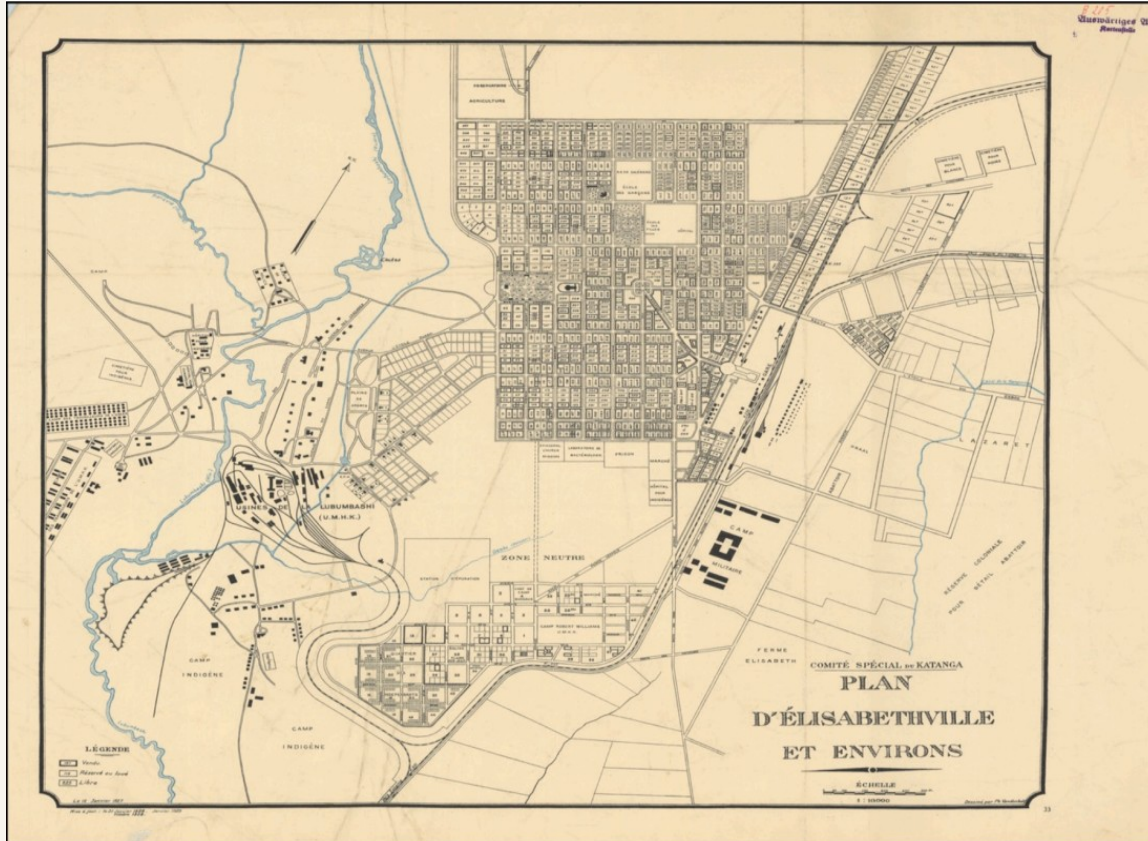
H2

An urban form characterized by horizontal spread, low verticality, and informal peri-urban settlements.

H3

Within-city inequality elevated by the segmented mining labor market and weak fiscal redistribution.

Colonial spatial logic — built before independence



Élisabethville urban plan, 1930.

Global commodities → local form

City founded to extract copper for global markets; mine geography dictated street geometry.

Coerced urbanization

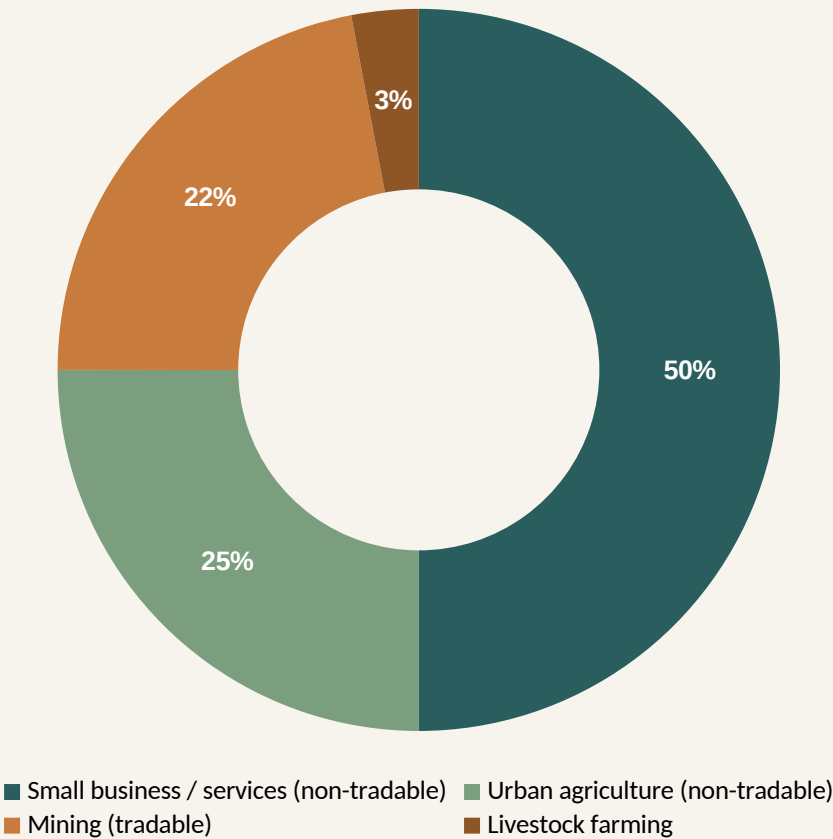
Belgian rule used taxation & forced labor to draw Africans from villages into mining-camp cities.

Segregation by design

European and African neighborhoods were spatially separated — a fragmentation that still shapes service access today.

75% of jobs are in non-tradables — and food is imported

Employment by sector (2022)



Food import dependence

Even basic foodstuffs come mostly from abroad — the textbook consumption-city profile.

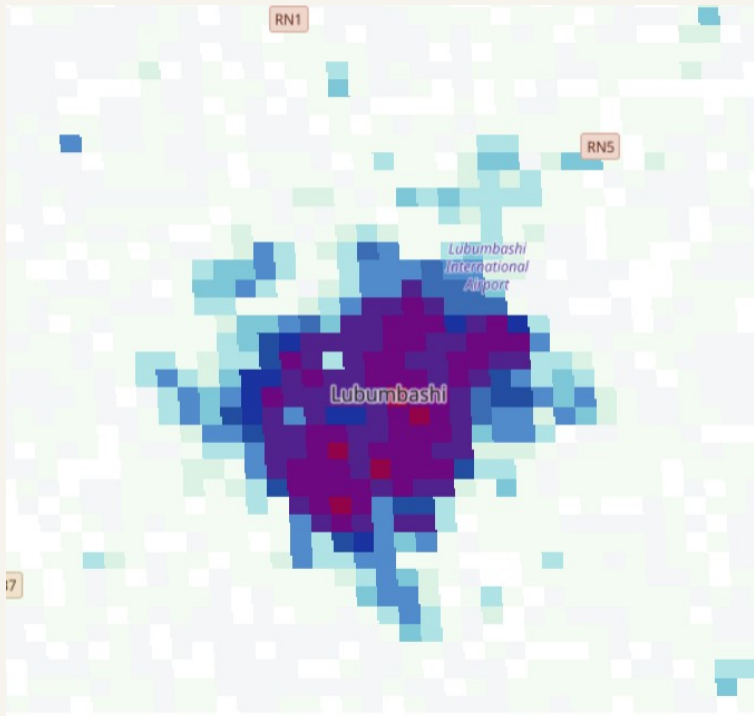
Imported share of urban food supply, Lubumbashi

Maize flour	44.1%
Rice	65.0%
Sugar	78.3%
Cooking oil	96.6%
Potatoes	94.7%
Fish & meat	55.0%

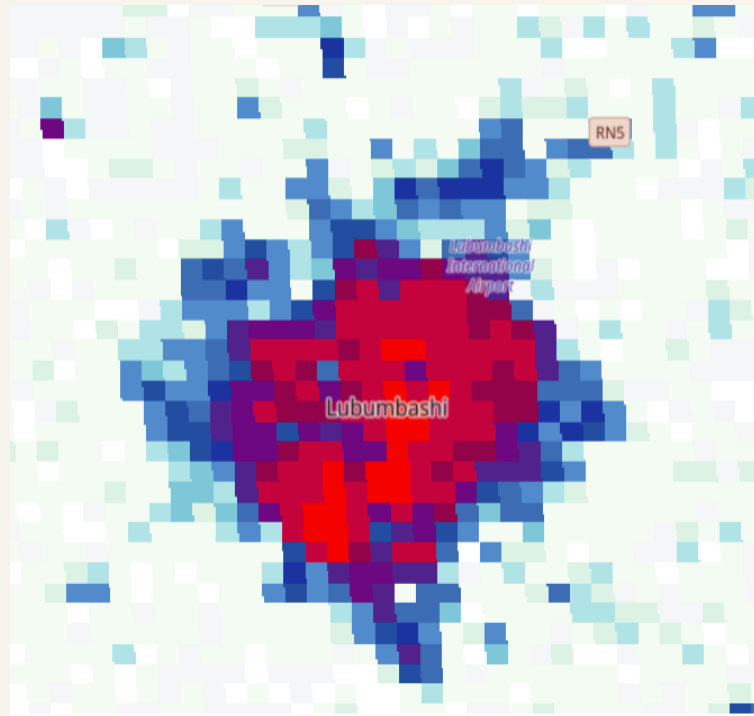
Sources: UN Industrial Development Organization (2022); Tshomba et al. (2020).

Forty years of horizontal spread

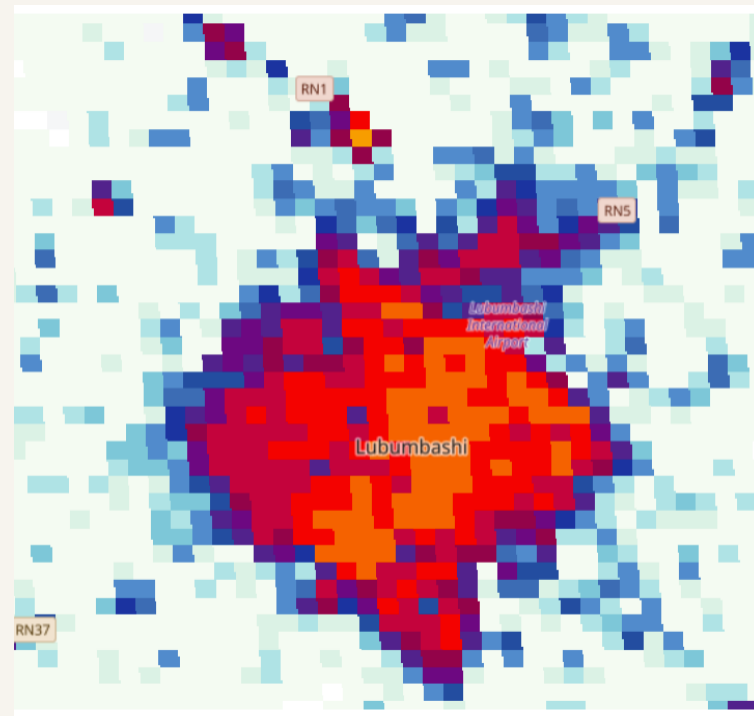
1985



2005



2025

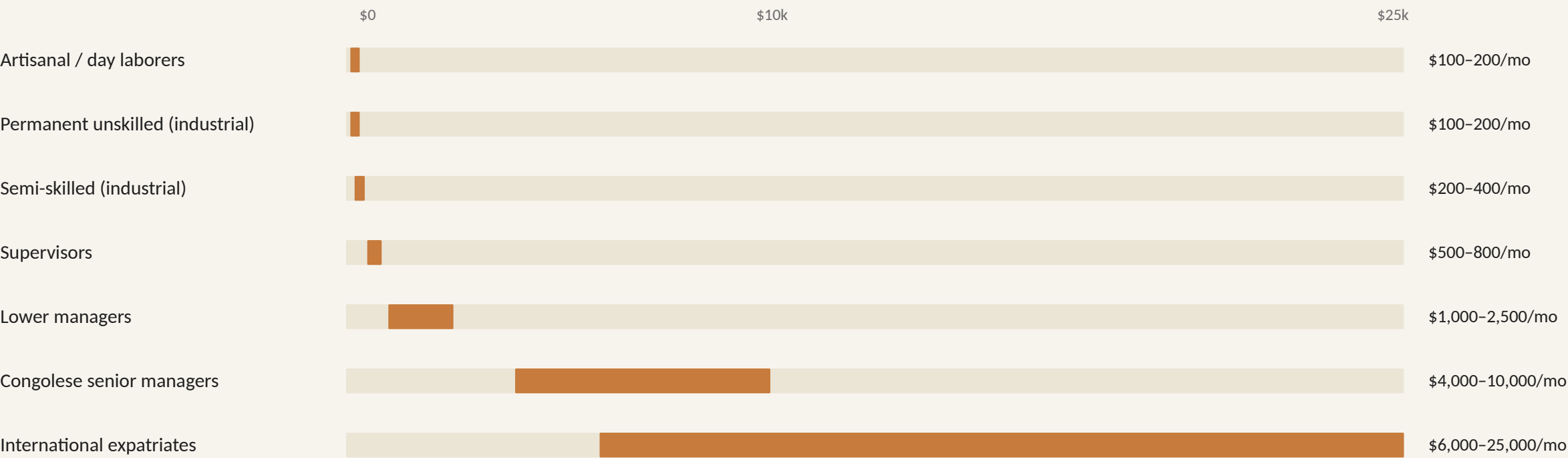


From a compact 1985 core, the city has sprawled outward four-fold — most growth is 4–20 km from the centre, along the NE corridor toward the airport and mines.

Source: Global Human Settlement Layer (EU); Sikuzani et al. (2018).

A steep, segmented wage ladder

Lubumbashi's mining labor market is sharply stratified — from artisanal day-laborers to expatriate managers earning 100× more.



Inequality channel: selective in-migration of better-educated workers lifts averages but widens local Gini — the pattern Loayza & Rigolini (2016) document across Peruvian mining districts.

Sources: Enquête 1-2-3 (Kankwanda et al., 2014); Rubbers (2020), Development & Change.

The fenceline: a concentrated human cost

99%

of fishers & farmers reported sharp yield declines from polluted rivers/wetlands

75%

couldn't afford healthcare or medicine for exposure-related illness

72%

reported recurring skin diseases after contact with contaminated water

60%

withdrew children from school due to lost income or illness

59%

reduced household food intake to one meal per day

56%

said women and girls are disproportionately affected — gynecological issues, birth defects

Survey of 144 respondents in copper-cobalt fenceline communities adjacent to industrial mines.
Source: RAID & AFREWATCH (2024), Environmental Pollution & the Human Costs of DRC's Cobalt Boom.

Hypotheses confirmed: Lubumbashi fits the model

HYPOTHESIS	LUBUMBASHI EVIDENCE	VERDICT
H1 — Non-tradables dominance + import dependence	~75% in non-tradables; 65–97% food import shares	✓ Confirmed
H2 — Horizontal spread, low verticality, peri-urban informality	4× sprawl since 1985; 3–5 m typical building heights	✓ Confirmed
H3 — Wide within-city inequality from mining labor segmentation	100× wage spread; fenceline livelihood collapse	✓ Confirmed

Mining reliance has shaped Lubumbashi's urban form to closely resemble the prototypical 'Consumption City' — with the social and spatial costs the literature predicts.

From extraction to urban productivity

If governance channels rents into city-wide public goods, the 'consumption-city' trap can be re-routed. Two complementary tracks:

Structural reforms

2018 Mining Code

Higher royalties + mandated Congolese ownership stakes.

Revenue redistribution

Direct mining royalties into fenceline water, sanitation, schools.

Environmental enforcement

Binding remediation standards; independent monitoring (Univ. of Lubumbashi).

Supply-chain accountability

EV-maker due diligence on community impacts (Tesla, VW, GM).

Urban planning responses

Buffer-zone planning

Enforce minimum residential setbacks; relocate most-exposed households.

Clean water access

Extend piped networks to peri-urban areas — only 11% of DRC has safe in-premise water.

Community grievance mechanisms

Formalize the 3+ active local court cases into a system; require company response.

Gender-responsive programs

Targeted health services for women in artisanal zones (Fair Cobalt Alliance model).

THANK YOU

A Consumption City, written by mining

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Abdul Sande